

L9: Lesson Summary

This lesson introduced you to several important points about epidemics on networks:

1. Compartmental epidemic models such as SIS and SIR under the homogeneous mixing assumption
2. The basic reproductive number and how it can be misleading in the presence of super-spreaders
3. Epidemic threshold
4. Real-world contact networks do not follow the homogeneous mixing assumption
5. How the spread of an epidemic depends on the second moment of the degree distribution
6. Power-law networks with diverging degree variance do not have an epidemic threshold
7. Computational modeling of epidemics



We will continue our study of spreading processes on networks in the next Lesson, considering some more advanced topics about epidemics as well as the spread of other entities on networks such as information and memes.